

3

The Polyakov path integral

We now begin a systematic study of string theory using the Polyakov path integral and conformal field theory. After an introduction to the path integral picture, we discuss gauge fixing, the Weyl anomaly, and vertex operators. We then generalize to strings propagating in curved spacetime.

3.1 Sums over world-sheets

The Feynman path integral is one way to represent a quantum theory, and it is a very natural method for describing interactions in string theory. In path integral quantization, amplitudes are given by summing over all possible histories interpolating between the initial and final states. Each history is weighted by

$$\exp(iS_{\text{cl}}/\hbar) , \tag{3.1.1}$$

with S_{cl} the classical action for the given history. Thus, one would define an amplitude in string theory by summing over all world-sheets connecting given initial and final curves, as in figure 3.1(a) for the open string and figure 3.1(b) for the closed string. The analogous sum for the relativistic point particle produces the free propagator (exercise 5.1).

One can imagine a number of ways in which strings might interact. One would be a contact interaction, an energy when two strings intersect. Another would be a long-range force mediated by some quantum field. However, it is not possible to *add* such interactions to string theory in a way that is consistent with the symmetries; we will get some idea later of why this is so. Rather, the only interactions that are allowed are those that are already implicit in the sum over world-sheets. Consider, for example, the world-sheets shown in figure 3.2. Figure 3.2(a) looks like a quantum correction to the open string amplitude of figure 3.1(a), involving intermediate states with two open strings. Figure 3.2(b) has three external

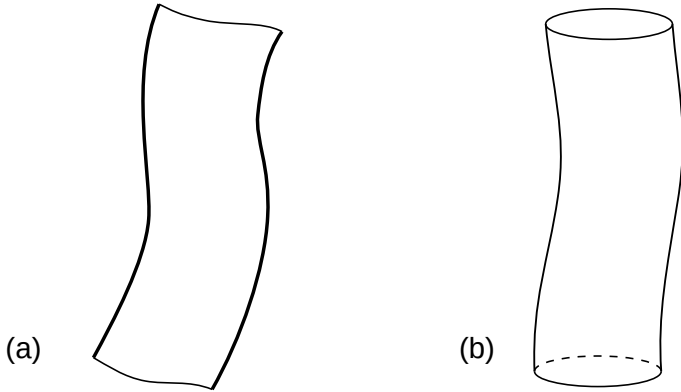


Fig. 3.1. (a) An open string world-sheet with the topology of a strip. The heavier curves are the world-lines of string endpoints. (b) A closed string world-sheet with the topology of a cylinder.

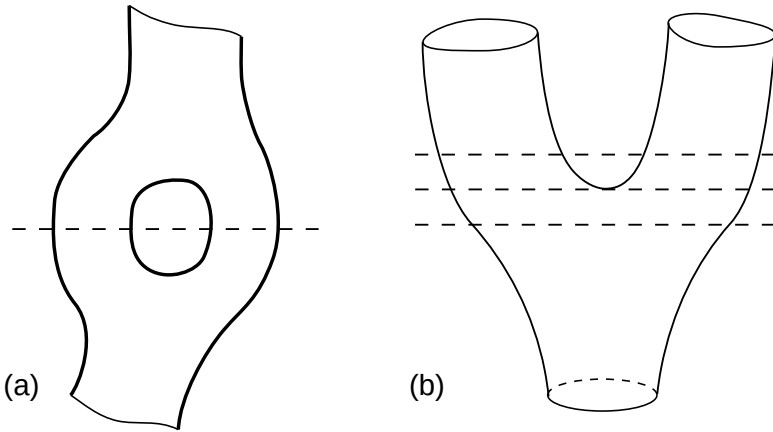


Fig. 3.2. (a) Quantum correction to open string propagation. (b) Decay of one closed string into two. The dashed lines are time-slices.

closed strings and represents one string decaying into two. It turns out that this is the correct way to introduce interactions into string theory. We will see that these interactions produce a consistent theory, finite and unitary, which includes gravity.

It is interesting to consider the process of figure 3.2(b) as seen at successive times as shown in figure 3.3. A closed string splits into two, or in the reverse process two join into one. This is the basic closed string interaction. In closed string theory, all particles are obtained as various states of excitation of the string, and all interactions (gauge, gravitational,

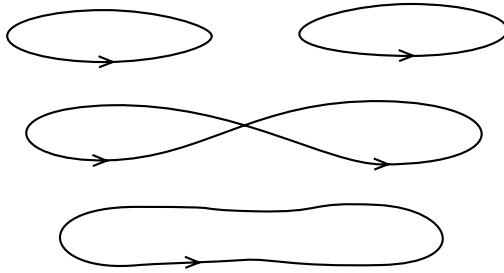
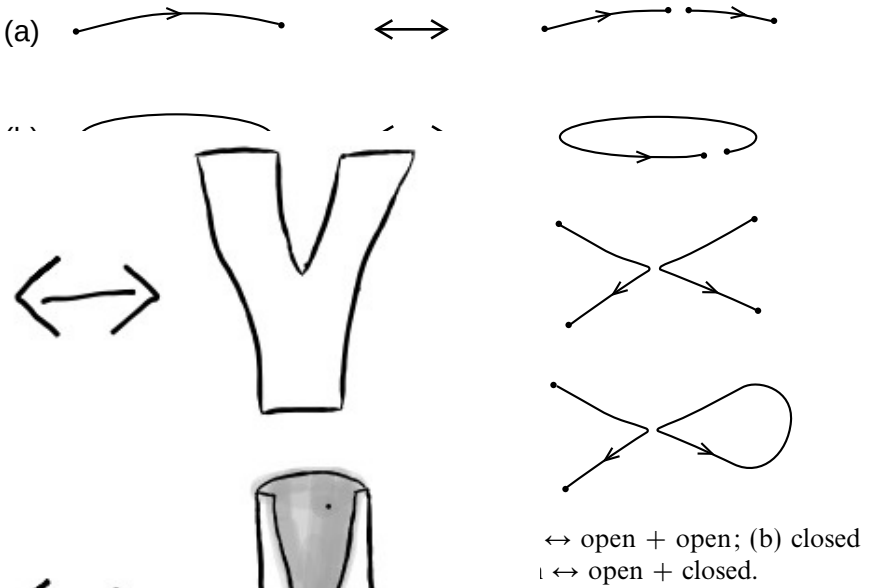


Fig. 3.3. Successive time-slices from figure 3.2(b). The arrows indicate the orientation of the string.



process of figure 3.3. For additional processes that can occur, the interactions of spacetime point, but this Lorentz frame, will put the distinguished point in the global topology of the world-sheet are the same. The introduction, it is this short-distance divergences

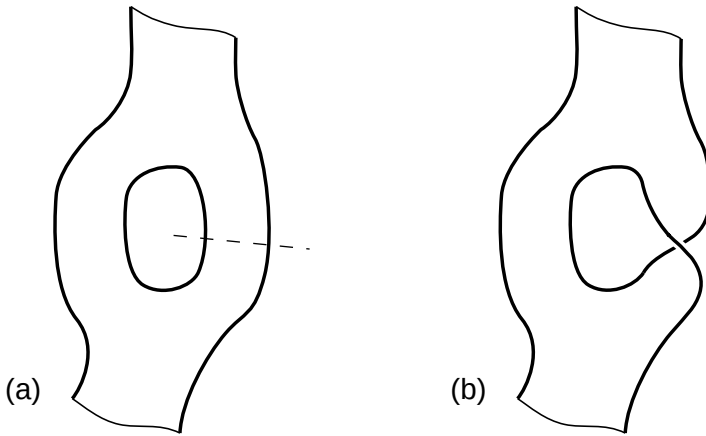


Fig. 3.5. Oriented (a) and unoriented (b) contributions to the two-open-string amplitude.

There are actually several different string theories, depending on which topologies we include in the sum over world-sheets. First note that the world-sheets we have drawn have two kinds of boundary, the source boundaries corresponding to the initial and final string configurations, and the endpoint boundaries corresponding to the world-lines of the endpoints of open strings. Ignore for now the source boundaries; we will discuss the details of the sources later in this chapter. So in the following discussion ‘boundary’ means endpoint boundary. There are then four ways to define the sum over world-sheets, which correspond to the four free string theories listed at the end of section 1.4:

1. Closed oriented: All oriented world-sheets without boundary.
2. Closed unoriented: All world-sheets without boundary.
3. Closed plus open oriented: All oriented world-sheets with any number of boundaries.
4. Closed plus open unoriented: All world-sheets with any number of boundaries.

There are two things that we should expand upon here. The first is the connection between the inclusion of unoriented world-sheets and the $\Omega = +1$ projection described in section 1.4. Figure 3.5 shows two world-sheets that are included in the unoriented theory with open strings. The unoriented world-sheet of figure 3.5(b) is equivalent to cutting the oriented surface of figure 3.5(a) along the dashed line, which we will parameterize

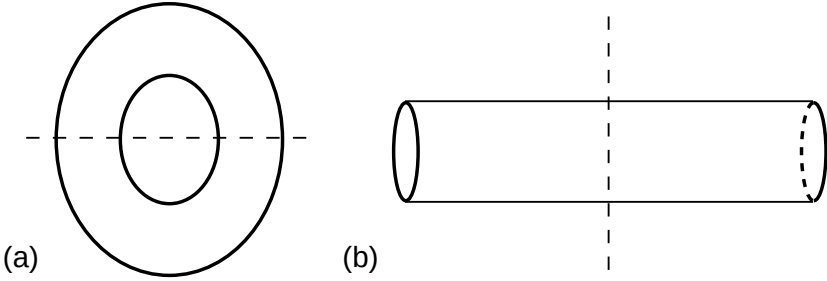


Fig. 3.6. (a) An annulus: the intermediate state is two open strings. (b) The same topology drawn as a cylinder: the intermediate state is one closed string.

by $0 \leq \sigma \leq \pi$, and requiring the fields on the cut edges to satisfy

$$X_{\text{upper}}^{\mu}(\sigma) = X_{\text{lower}}^{\mu}(\pi - \sigma) . \quad (3.1.2)$$

This in turn is equivalent to acting with the operator Ω on the open string state. Adding these two surfaces with appropriate weight then amounts to inserting the operator $\frac{1}{2}(1 + \Omega)$, which projects onto states of $\Omega = +1$. Summing over all oriented and unoriented surfaces inserts projection operators on all intermediate states, reducing the spectrum to the $\Omega = +1$ sector.

Second, the list above does not include any theory with open strings but not closed. Let us explain in more detail why this is the case. Consider a world-sheet with the topology of an annulus, figure 3.6(a). For convenience we have drawn only the vacuum amplitude, but by attaching appropriate sources the same argument will also apply to scattering amplitudes. We see from the dashed line that this is a process with an intermediate state of two open strings, and so world-sheets of this topology must be present in any theory with open strings. In figure 3.6(b) we have drawn the same topology as a cylinder and cut it open on an intermediate state of a single closed string. Thus, even if we start with open strings only, the sum over world-sheets will necessarily include processes in which scattering of open strings produces closed strings. As we develop the amplitudes we will see how this works in detail.

The need for closed strings in open string theory can also be seen in another way. Consider the interactions in figures 3.4(a) and (b), but time reversed: respectively two open strings joining into one, and an open to closed string transition. Near the interaction point these are the same, two endpoints coalescing. To have the first interaction without the second would require some nonlocal constraint on the dynamics of the string; this would surely be inconsistent. One can make the same argument with figures 3.4(c) and (d), where the interaction is locally the reconnection of